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CBSE CLASS 10 | MATHEMATICS

ARITHMETIC PROGRESSIONS

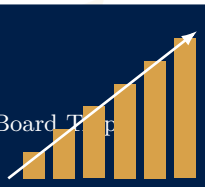
Master Concept, Formula & Visual Problem-Solving Handbook

Academic Year 2026–27

Every constant step builds infinity.
Master the discrete mathematics of linear patterns.

General Term a_n | Terminal Mechanics | Gauss's Sum S_n | The Bridge $S_n \leftrightarrow a_n$ | Board Tr...p

A student-first Maths Manthan conceptual and visual revision resource



What This Handbook Includes

- Sequence vs Series vs Arithmetic Progression
- General term derivation & the fencepost principle
- Terminal mechanics: n-th term from end & middle terms
- Arithmetic mean & symmetric term selection techniques
- Gauss's forward-backward summation proof of S_n
- The telescoping bridge: $a_n = S_n - S_{n-1}$ & degree hacks
- Double answer mechanics in decreasing series
- CBSE Section-E contextual case study archetypes

Pedagogical Strategy: Foundations First

Arithmetic Progression is the discrete counterpart of the continuous linear function $y = mx + c$. The common difference d represents constant slope, while n is strictly quantized to positive integers:

$$\mathbf{a_n = d \cdot n + (a - d)} \longleftrightarrow \mathbf{y = mx + c}$$

$$\mathbf{S_n = \left(\frac{d}{2}\right) n^2 + \left(a - \frac{d}{2}\right) n} \longleftrightarrow \text{Linear Area}$$

Always state formulas with domain conditions: $\mathbf{n \in \{1, 2, 3, \dots\}}$.

CBSE 2026–27 Curriculum Verification & Rationalized Scope

Official Scope (Prescribed & Tested): Motivation for studying AP, derivation of the n-th term ($a_n = a + (n-1)d$) and sum of first n terms ($S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}[a + l]$), their application in solving daily life contextual problems.

Excluded / Deleted Topics (Non-Examinable):

- **Geometric Progressions (GP) & Harmonic Progressions (HP):** Strictly outside Class 10 scope (covered in Class 11).
- **Sum of Squares ($\sum n^2$) and Sum of Cubes ($\sum n^3$):** Strictly non-examinable in Class 10.
- **Infinite Arithmetic Progressions Summation:** Only finite sums S_n are evaluated.
- **Non-linear daily-life word problems:** Complex multi-equation systems not reducible to linear AP relations.

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Prepared for learning and conceptual mastery – Strictly aligned with CBSE Class 10 Syllabus 2026–27.

CHAPTER 05

CORE SYLLABUS

ARITHMETIC PROGRESSIONS

Module 1 | Foundational Anatomy: Sequences, Series & Common Difference

1.1 Hierarchy: Sequence, Series, and Progression

- **Sequence:** An ordered arrangement of numbers according to a definite rule: $a_1, a_2, a_3, \ldots, a_n$. Each element is a **term**.
- **Series:** The indicated sum of the terms of a sequence: $a_1 + a_2 + a_3 + \cdots + a_n$.
- **Arithmetic Progression (AP):** A sequence where each term after the first is obtained by adding a fixed constant d to the preceding term.

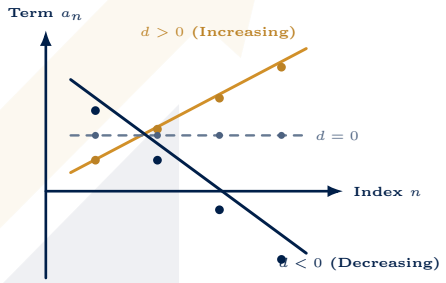
Formal Definition of an AP:

$$\mathbf{a_{k+1} - a_k = d} \quad \text{for all } \mathbf{k} \in \mathbb{N}$$

The Three Regimes of Common Difference ($d \in \mathbb{R}$):

- $d > 0$ **(Increasing AP):** $a_1 < a_2 < a_3 < \ldots$
  e.g., $3, 7, 11, 15, \ldots$ $(d = +4)$.
- $d < 0$ **(Decreasing AP):** $a_1 > a_2 > a_3 > \ldots$
  e.g., $20, 15, 10, 5, \ldots$ $(d = -5)$.
- $d = 0$ **(Stationary AP):** $a_1 = a_2 = a_3 = \ldots$
  e.g., $7, 7, 7, 7, \ldots$ $(d = 0)$.

Note: d can be rational, irrational, positive, or negative.



1.2 Deterministic Verification Protocol

Algorithmic AP Verification Protocol

- **Explicit List of Terms ($a_1, a_2, a_3, \ldots$):** Calculate consecutive differences $d_1 = a_2 - a_1, d_2 = a_3 - a_2, \ldots$. The sequence is an AP if and only if $d_1 = d_2 = d_3 = \cdots = d$ (constant).
- **Algebraic Formula (a_n):** Compute a_{n+1} and evaluate $\Delta_n = a_{n+1} - a_n$. If Δ_n is a constant independent of n, it is an AP with common difference $d = \Delta_n$.

Worked Archetype 1 | Algebraic and Radical AP Verification

Problem: (i) Is $a_n = 4n + 7$ an AP? (ii) Do $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, \ldots$ form an AP?
Solution:

1. $a_{n+1} = 4(n+1) + 7 = 4n + 11 \implies a_{n+1} - a_n = (4n + 11) - (4n + 7) = \mathbf{4}$ (Constant, independent of n).
   $\therefore$ The sequence is an AP with common difference $d = 4$ and first term $a_1 = 4(1) + 7 = 11$.
2. Simplify radicals: $a_1 = \sqrt{2}, a_2 = 2\sqrt{2}, a_3 = 3\sqrt{2}, a_4 = 4\sqrt{2}$.
   Consecutive differences: $a_2 - a_1 = \sqrt{2}, a_3 - a_2 = \sqrt{2}, a_4 - a_3 = \sqrt{2}$.
   $\because a_{k+1} - a_k = \sqrt{2} =$ constant, the sequence **forms an AP with** $a = \sqrt{2}$ **and** $d = \sqrt{2}$.

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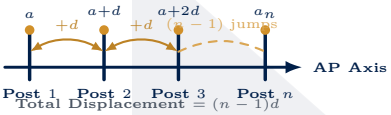
Module 2 | The General Term: Inductive Derivation & Fencepost Geometry

2.1 Inductive Derivation & The Fencepost Principle

Let an AP have first term a and common difference d. Successive terms unfold inductively:

$$a_1 = a = a + (1-1)d, \quad a_2 = a + 1d = a + (2-1)d, \quad a_3 = a + 2d = a + (3-1)d$$

By mathematical induction, the coefficient of d is always 1 less than the position index:



The General Term (n-th Term) Master Formula:

$$\mathbf{a_n = a + (n-1)d}$$

- $\mathbf{a}$ = First term of the progression ($a = a_1$)
- $\mathbf{d}$ = Common difference ($d = a_{k+1} - a_k$)
- $\mathbf{n}$ = Term position ($\mathbf{n} \in \mathbb{N} = \{1, 2, 3, \dots\}$)
- $\mathbf{a_n}$ = Value of the n-th term (or last term l)

The Fencepost Rule: To build n posts, exactly $(n-1)$ gaps are spanned. Hence $(n-1)$ additions of d reach term a_n.

2.2 The Linear Function Signature ($a_n = An + B$)

Expanding the general term algebraically:

$$a_n = a + nd - d = \mathbf{d \cdot n + (a - d)}$$

Linear Theorem of AP: A sequence is an Arithmetic Progression if and only if its n-th term is a **linear polynomial in** n:

$$\mathbf{a_n = An + B} \implies \text{Common Difference } \mathbf{d = A}, \quad \text{First Term } \mathbf{a = A + B}$$

MCQ Rule: The coefficient of n is **always** the common difference d!

Worked Archetype 2 | Position Membership and Domain Verification

Problem: (i) Which term of the AP: $3, 8, 13, 18, \dots$ is 78? (ii) Is 301 a term of $5, 11, 17, 23, \dots$?
Solution:

1. Here, $a = 3$, $d = 8 - 3 = 5$. Set $a_n = 78$:

$$3 + (n-1)5 = 78 \implies 5(n-1) = 75 \implies n - 1 = 15 \implies \mathbf{n = 16}$$

   $\therefore$ The 16**-th term** of the AP is 78.
2. Here, $a = 5$, $d = 11 - 5 = 6$. Set $a_n = 301$:

$$5 + (n-1)6 = 301 \implies 6(n-1) = 296 \implies n - 1 = \frac{148}{3} \implies n = \frac{\mathbf{151}}{\mathbf{3}} = \mathbf{50\frac{1}{3}}$$

Conclusion: Since position n must be a natural number ($n \in \mathbb{N}$) and $\frac{151}{3} \notin \mathbb{N}$, 301 **is NOT a term**.

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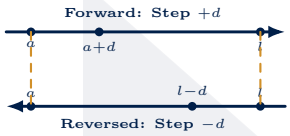
ARITHMETIC PROGRESSIONS

Module 3 | Terminal Mechanics: n-th Term from the End & Middle Term Protocols

3.1 The n-th Term from the End of a Finite AP

Consider a finite AP with first term a, common difference d, and last term l:

$$a,\ a+d,\ a+2d,\ \ldots,\ l-2d,\ l-d,\ l$$



Master Formula: n-th Term from the End (a'_n):

$$\mathbf{a'_n = l - (n-1)d}$$

Invariance Property: The sum of the n-th term from the beginning and the n-th term from the end is **constant**:

$$\mathbf{a_n + a'_n = a + l}$$

Position Equivalence: In an AP of m total terms, the n-th term from the end is the $(m-n+1)$**-th term from the start**.

3.2 Middle Term(s) Protocol for Finite APs

To determine the middle term of an AP $a, a+d, \ldots, l$, first calculate total terms: $\mathbf{m = \frac{l-a}{d} + 1}$.

Total Terms (m)	Number of Middle Terms	Exact Position & Formula
Case 1: m is ODD	Exactly **ONE** middle term	Position: $\mathbf{k = \frac{m+1}{2}}$ Value: $\mathbf{a_k = a + \left(\frac{m-1}{2}\right) d}$
Case 2: m is EVEN	Exactly **TWO** middle terms	Positions: $\mathbf{k_1 = \frac{m}{2}}$ and $\mathbf{k_2 = \frac{m}{2} + 1}$

Worked Archetype 3 | End-Term and Middle-Term Identification

Problem: Given the AP: $10, 7, 4, \ldots, -62$. (i) Find the 11-th term from the end. (ii) Find the middle term of the AP.

Solution:

1. Here, $a = 10$, $d = 7 - 10 = -3$, and last term $l = -62$. Using the end-term formula for $n = 11$:

$$a'_{11} = l - (11-1)d = -62 - (10)(-3) = -62 + 30 = \mathbf{-32}$$

2. First find total terms m: $-62 = 10 + (m-1)(-3) \implies -72 = -3(m-1) \implies m - 1 = 24 \implies m = 25$. Since $m = 25$ is **ODD**, there is exactly one middle term at position:

$$k = \frac{m+1}{2} = \frac{25+1}{2} = \mathbf{13}$$

Value of middle term $a_{13} = a + 12d = 10 + 12(-3) = 10 - 36 = \mathbf{-26}$.

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Module 4 | Arithmetic Mean & Symmetrical Selection of Terms

4.1 The Three-Term Equilibrium Condition & Arithmetic Mean

If three quantities a, b, c are in AP, the common difference is identical:

$$b - a = c - b \implies \mathbf{2b = a + c} \implies \mathbf{b = \frac{a + c}{2}}$$



Symmetric Balance around a

Symmetrical Variable Selection Protocol:

- **3 Terms:** $\mathbf{a-d},\ \mathbf{a},\ \mathbf{a+d}$ (Common Diff = **d**, Sum = **3a**)
- **4 Terms:** $\mathbf{a-3d},\ \mathbf{a-d},\ \mathbf{a+d},\ \mathbf{a+3d}$ (Common Diff = **2d**

  TRAP ALERT

  , Sum = **4a**)
- **5 Terms:** $\mathbf{a-2d},\ \mathbf{a-d},\ \mathbf{a},\ \mathbf{a+d},\ \mathbf{a+2d}$ (Common Diff = **d**, Sum = **5a**)

Equilibrium: Symmetrical terms cause d to cancel completely upon summing!

4.2 The 4-Term Common Difference Trap

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The 4-Term Common Difference Trap! In the 4-term selection $(a-3d, a-d, a+d, a+3d)$, the difference between consecutive terms is $(a-d)-(a-3d) = \mathbf{2d}$ (NOT **d**!). If you solve and get $d = 2$, the actual step of the AP is $\mathbf{2d = 4}$.

Worked Archetype 4 | 3 Symmetrical Numbers in AP

Problem: The sum of three numbers in an AP is 27 and their product is 648. Find the numbers.
Solution:

1. Let the three numbers in AP be $\mathbf{a-d},\ \mathbf{a},\ \mathbf{a+d}$.
2. Sum condition: $(a-d)+a+(a+d) = 27 \implies 3a = 27 \implies \mathbf{a = 9}$.
3. Product condition: $(a-d)\cdot a\cdot(a+d) = 648 \implies 9(81-d^2) = 648 \implies 81-d^2 = 72 \implies d^2 = 9 \implies \mathbf{d = \pm 3}$.
4. For $d = 3$, numbers are $9-3, 9, 9+3 \implies \mathbf{6, 9, 12}$. For $d = -3$, numbers are $\mathbf{12, 9, 6}$.
   $\therefore$ The required numbers are $6, 9, 12$.

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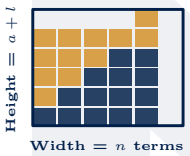
Module 5 | The Sum of First n Terms: Gauss's Proof & Dual Formulas

5.1 Gauss's Discovery & The Forward-Backward Columnar Proof

Writing the series forwards and backwards and adding column-wise:

$$S_n = a + (a+d) + (a+2d) + \cdots + l$$
$$S_n = l + (l-d) + (l-2d) + \cdots + a$$
$$2S_n = \underbrace{(a+l) + (a+l) + \cdots + (a+l)}_{n \text{ identical pairs}} = n(a+l) \implies \mathbf{S_n = \frac{n}{2}(a+l)}$$

Substituting $l = a + (n-1)d$ yields: $\mathbf{S_n = \frac{n}{2}[2a + (n-1)d]}$.



The Dual Master Sum Formulas:

$$\mathbf{S_n = \frac{n}{2}[2a + (n-1)d]} \quad \longleftrightarrow \quad \mathbf{S_n = \frac{n}{2}[a+l]}$$

Standard Natural Number Summations:

- First n natural numbers: $\sum_{k=1}^{n} \mathbf{k} = \frac{\mathbf{n(n+1)}}{\mathbf{2}}$
- First n odd naturals: $\mathbf{1 + 3 + \cdots + (2n-1) = n^2}$
- First n even naturals: $\mathbf{2 + 4 + \cdots + 2n = n(n+1)}$

5.2 Direct Sum Applications

Worked Archetype 5 | Direct Application of Sum Formulas

Problem: Find the sum of: (i) first 22 terms of $8, 3, -2, \ldots$ (ii) the series $34 + 32 + 30 + \cdots + 10$.
Solution:

1. Here, $a = 8$, $d = 3 - 8 = -5$, and $n = 22$.

$$S_{22} = \frac{22}{2}[2(8) + (22-1)(-5)] = 11[16 + 21(-5)] = 11[16 - 105] = 11(-89) = \mathbf{-979}$$

2. Here, $a = 34$, $d = 32 - 34 = -2$, and last term $l = 10$. First find n:

$$10 = 34 + (n-1)(-2) \implies -24 = -2(n-1) \implies n - 1 = 12 \implies \mathbf{n = 13}$$

Using the last-term formula: $S_{13} = \frac{13}{2}[a + l] = \frac{13}{2}[34 + 10] = \frac{13}{2}[44] = 13 \times 22 = \mathbf{286}$.

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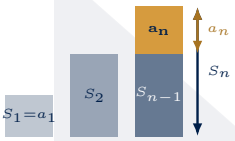
ARITHMETIC PROGRESSIONS

Module 6 | The Bridge Principle: Connecting S_n and a_n & Polynomial Degrees

6.1 The Telescoping Difference Identity

Subtracting consecutive cumulative sums isolates the individual n-th term:

$$S_n - S_{n-1} = (a_1 + \cdots + a_{n-1} + a_n) - (a_1 + \cdots + a_{n-1}) = \mathbf{a_n} \quad (\mathbf{n \geq 2})$$



The Bridge Formula:

$$\mathbf{a_n = S_n - S_{n-1}} \quad (\mathbf{n \geq 2}), \quad \mathbf{a_1 = S_1}$$

The Quadratic Signature of S_n:

$$S_n = \left(\frac{d}{2}\right) n^2 + \left(a - \frac{d}{2}\right) n = \mathbf{An^2 + Bn}$$

Speed Hacks for Board MCQs:

- Common Difference: $\mathbf{d = 2A}$ (Twice coefficient of n^2)
- First Term: $\mathbf{a = A + B = S_1}$
- Constant Term: Must be **zero** for an AP from $n = 1$.

6.2 Board Examination Protocol for $a_n = S_n - S_{n-1}$

Worked Archetype 6 | Step-by-Step Derivation of a_n from S_n

Problem: The sum of the first n terms of an AP is $S_n = 3n^2 + 5n$. Find the n-th term and the 15-th term.
Solution:

1. **Express S_{n-1}:** Replace n with $(n-1)$:

$$S_{n-1} = 3(n-1)^2 + 5(n-1) = 3(n^2 - 2n + 1) + 5n - 5 = 3n^2 - 6n + 3 + 5n - 5 = \mathbf{3n^2 - n - 2}$$

2. **Apply Bridge Identity:**

$$a_n = S_n - S_{n-1} = (3n^2 + 5n) - (3n^2 - n - 2) = 3n^2 + 5n - 3n^2 + n + 2 = \mathbf{6n + 2}$$

3. **Boundary Check:** $a_1 = 6(1) + 2 = 8$. From formula, $S_1 = 3(1)^2 + 5(1) = 8$. Verified!
4. **Find 15th Term:** $a_{15} = 6(15) + 2 = 90 + 2 = \mathbf{92}$.
   Verification Hack: $S_n = 3n^2 + 5n \implies A = 3, B = 5 \implies d = 2(3) = 6, a = 3 + 5 = 8 \implies a_n = 6n + 2$.

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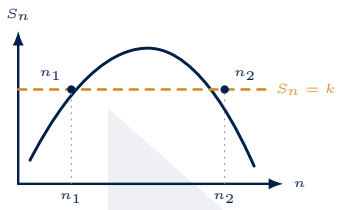
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Module 7 | The Dual Roots of $S_n = k$: Explaining Double Answers

7.1 The "Double Answer" Mechanism in Board Examinations

When solving $S_n = k$, quadratic factorization frequently yields **two distinct natural numbers** $n = n_1$ and $n = n_2$ $(n_2 > n_1)$.



The Null-Sum Block Condition: Two term counts n_1 and n_2 produce identical sums $S_{n_1} = S_{n_2} = k$ if and only if:

$$\sum_{\mathbf{j=n_1+1}}^{\mathbf{n_2}} \mathbf{a_j = 0}$$

Mechanism: Occurs when $a > 0$ and $d < 0$. The terms cross zero into negative values, and the negative terms cancel out the positive tail.

7.2 Step-by-Step Classic Board Archetype

Worked Archetype 7 | The Classic Double Root Board Model

Problem: How many terms of the AP: $24, 21, 18, \ldots$ must be taken so that their sum is 78? Explain the double answer.

Solution:

1. Here, $a = 24$, $d = 21 - 24 = -3$, and $S_n = 78$.
2. Sum equation:

$$78 = \frac{n}{2}[2(24) + (n-1)(-3)] \implies 156 = n[48 - 3n + 3] = n[51 - 3n] \implies 3n^2 - 51n + 156 = 0$$

3. Dividing by 3: $n^2 - 17n + 52 = 0 \implies (n-4)(n-13) = 0 \implies \mathbf{n = 4}$ or $\mathbf{n = 13}$.
4. **Board Explanation of Double Answer:** Both $n = 4$ and $n = 13$ are positive integers. The first 4 terms sum to $24 + 21 + 18 + 15 = 78$. The remaining terms from 5-th to 13-th are:

$$12 + 9 + 6 + 3 + 0 + (-3) + (-6) + (-9) + (-12) = \mathbf{0}$$

Because this block sums to zero, adding them leaves the sum unchanged. Thus, **both answers are valid**.

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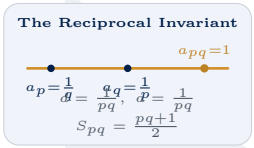
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Module 8 | Parameterized AP Models: p-th, q-th Terms & Unknown Constants

8.1 Universal Board Examination Identities

Given Hypothesis	Deduced Parameters (a, d)	Universal Board Result
$\mathbf{a_p = q}$ and $\mathbf{a_q = p}$	$d = -1, \quad a = p + q - 1$	$\mathbf{a_{p+q} = 0}$
$\mathbf{p \cdot a_p = q \cdot a_q} \quad (p \neq q)$	$a + (p + q - 1)d = 0$	$\mathbf{a_{p+q} = 0}$
$\mathbf{a_p = \frac{1}{q}}$ and $\mathbf{a_q = \frac{1}{p}}$	$a = \frac{1}{pq}, \quad d = \frac{1}{pq}$	$\mathbf{a_{pq} = 1}, \quad \mathbf{S_{pq} = \frac{1}{2}(pq + 1)}$



Protocol for Unknown Variables: If $f(x), g(x), h(x)$ are in AP:

$$\mathbf{2 \cdot g(x) = f(x) + h(x)}$$

$$\iff \mathbf{g(x) - f(x) = h(x) - g(x)}$$

Always equate consecutive differences directly!

8.2 Worked Algebraic Parameter Archetype

Worked Archetype 8 | Unknown Parameter Determination

Problem: (i) If $a_p = \frac{1}{q}$ and $a_q = \frac{1}{p}$, prove $S_{pq} = \frac{1}{2}(pq + 1)$. (ii) Find x such that $(x + 2), 2x, (2x + 3)$ are in AP.
Solution:

1. State equations: $a + (p - 1)d = \frac{1}{q}$ and $a + (q - 1)d = \frac{1}{p}$. Subtracting:

$$(p - q)d = \frac{1}{q} - \frac{1}{p} = \frac{p - q}{pq} \implies \mathbf{d = \frac{1}{pq}}, \quad \mathbf{a = \frac{1}{pq}}$$

   Now evaluate S_{pq}: $S_{pq} = \frac{pq}{2}\left[\frac{2}{pq} + (pq - 1)\frac{1}{pq}\right] = \frac{pq}{2}\left[\frac{pq+1}{pq}\right] = \mathbf{\frac{pq+1}{2}}$.
2. For $(x + 2), 2x, (2x + 3)$ in AP: $2(2x) = (x + 2) + (2x + 3) \implies 4x = 3x + 5 \implies \mathbf{x = 5}$.

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Module 9 | The Master Ratio Transformation: Sum Ratios to Term Ratios

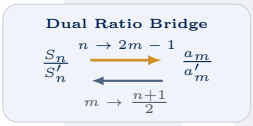
10.1 The Master Transformation Theorem

Given the ratio of sums of two arithmetic progressions:

$$\frac{\mathbf{S_n}}{\mathbf{S'_n}} = \frac{\mathbf{f(n)}}{\mathbf{g(n)}}$$

The ratio of their **m**-th terms $\frac{\mathbf{a_m}}{\mathbf{a'_m}}$ is obtained by replacing **n** with $(\mathbf{2m-1})$:

$$\frac{\mathbf{a_m}}{\mathbf{a'_m}} = \frac{\mathbf{f(2m-1)}}{\mathbf{g(2m-1)}}$$



Mathematical Proof of the $(2m-1)$ Hack:

$$\frac{S_n}{S'_n} = \frac{\frac{n}{2}[2a_1+(n-1)d_1]}{\frac{n}{2}[2a_2+(n-1)d_2]} = \frac{a_1+\left(\frac{n-1}{2}\right)d_1}{a_2+\left(\frac{n-1}{2}\right)d_2}$$

We require $\frac{a_1+(m-1)d_1}{a_2+(m-1)d_2}$. Equating coefficients:

$$\frac{n-1}{2} = m-1 \implies \mathbf{n = 2m-1}$$

10.2 High-Scoring Board Archetype

Worked Archetype 9 | Sum Ratio to Term Ratio Board Master

Problem: The ratio of the sums of n terms of two arithmetic progressions is $(7n+1) : (4n+27)$. Find the ratio of their 11-th terms.

Solution:

1. State sum ratio: $\frac{S_n}{S'_n} = \frac{2a_1+(n-1)d_1}{2a_2+(n-1)d_2} = \frac{a_1+\left(\frac{n-1}{2}\right)d_1}{a_2+\left(\frac{n-1}{2}\right)d_2} = \frac{7n+1}{4n+27}$.
2. We require the ratio of their 11-th terms: $\frac{a_{11}}{a'_{11}} = \frac{a_1+10d_1}{a_2+10d_2}$.
3. Set $\frac{n-1}{2} = 10 \implies n-1 = 20 \implies \mathbf{n = 21}$.
4. Substitute $n = 21$ into the ratio:

$$\frac{a_{11}}{a'_{11}} = \frac{7(21)+1}{4(21)+27} = \frac{147+1}{84+27} = \frac{148}{111} = \mathbf{\frac{4}{3}} \quad \text{(dividing by 37)}$$

$\therefore$ The ratio of their 11-th terms is $4 : 3$.

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QUICK REFERENCE

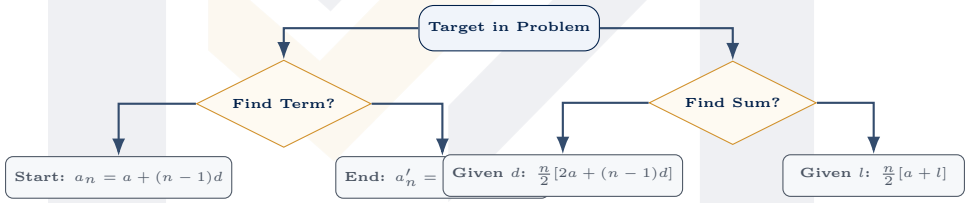
ARITHMETIC PROGRESSIONS

Module 10 | Master Formula Matrix & Problem-Solving Decision Engine

11.1 Comprehensive Master Formula Matrix

Target Concept	Master Formula	Key Operational Conditions
Common Difference	$d = a_{k+1} - a_k$	Constant for all $k \geq 1$; can be $> 0, < 0, = 0$.
General Term (n-th)	$a_n = a + (n-1)d$	$n \in \mathbb{N} = \{1, 2, 3, \dots\}$; linear polynomial $dn + (a-d)$.
n-th Term from End	$a'_n = l - (n-1)d$	l is last term; invariance: $a_n + a'_n = a + l$.
Middle Term(s)	$k = \frac{m+1}{2}$ (odd m)	1 term if m is odd; 2 terms $(\frac{m}{2}, \frac{m}{2}+1)$ if m is even.
Arithmetic Mean	$b = \frac{a+c}{2} \iff 2b = a + c$	For 3 terms a, b, c in AP.
Sum of n Terms (d)	$S_n = \frac{n}{2}[2a + (n-1)d]$	Quadratic in n with zero constant: $An^2 + Bn$.
Sum of n Terms (l)	$S_n = \frac{n}{2}[a + l]$	Deploy whenever last term $l = a_n$ is known.
Term from Sums	$a_n = S_n - S_{n-1}$ $(n \geq 2)$	Always verify $a_1 = S_1$. Common difference $d = 2A$.

11.2 Algorithmic Problem-Solving Decision Flowchart



11.3 Standard Board Examination Presentation Protocol

Golden Rules for Securing 100% Marks in Section C & D

- **Explicit Definitions:** Always state: *"Let a be the first term and d be the common difference of the AP."*
- **Symbolic Formulas First:** Always write the general formula in standard algebraic symbols before plugging in numbers. CBSE marking schemes reserve 1/2 mark specifically for the formula.
- **Integer Sanity Checks:** Whenever calculating n, explicitly state: *"Since $n \in \mathbb{N}$, $n = \dots$ is accepted / rejected."*

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EXAM STRATEGY

ARITHMETIC PROGRESSIONS

Module 11 | Worked Board Exam Master Archetypes (Step-by-Step Rigor)

12.1 Archetype A: The First Negative Term Protocol (Inequality Method)

Board Model | Finding the First Negative Term

Problem: Find the first negative term of the AP: $121, 117, 113, \ldots$
Solution:

1. Here, $a = 121$, $d = 117 - 121 = -4$. Set the strict inequality:

$$a_n < 0 \implies 121 + (n-1)(-4) < 0 \implies 121 - 4n + 4 < 0 \implies 125 < 4n \implies \mathbf{n > \frac{125}{4} = 31.25}$$

2. Since n must be the smallest natural number greater than 31.25, we take $\mathbf{n = 32}$.
3. **Value:** $a_{32} = 121 + 31(-4) = 121 - 124 = \mathbf{-3}$. The **32-nd term** is the first negative term.

12.2 Archetype B: Multiples & Divisibility in a Number Range

Board Model | Sum of Divisible Numbers in a Given Interval

Problem: Find the sum of all natural numbers between 100 and 500 which are divisible by 7.
Solution:

1. Boundaries: Strictly $101 \le x \le 499$. First term $a = 105$, last term $l = 497$, difference $d = 7$.
2. Number of terms: $497 = 105 + (n-1)7 \implies 392 = 7(n-1) \implies n - 1 = 56 \implies \mathbf{n = 57}$.
3. Total sum: $S_{57} = \frac{57}{2}[a + l] = \frac{57}{2}[105 + 497] = \frac{57}{2}[602] = 57 \times 301 = \mathbf{17157}$.

12.3 Archetype C: Symmetric 4-Term Ratio Model

Board Model | 4 Terms with Ratio of Extremes to Means

Problem: Divide 32 into four parts in AP such that the ratio of the product of extremes to the product of means is $7 : 15$.
Solution:

1. Let the parts be $\mathbf{a - 3d}$, $\mathbf{a - d}$, $\mathbf{a + d}$, $\mathbf{a + 3d}$ (Common difference $= 2d$).
2. Sum: $(a - 3d) + (a - d) + (a + d) + (a + 3d) = 32 \implies 4a = 32 \implies \mathbf{a = 8}$.
3. Ratio: $\frac{(8-3d)(8+3d)}{(8-d)(8+d)} = \frac{64-9d^2}{64-d^2} = \frac{7}{15} \implies 15(64 - 9d^2) = 7(64 - d^2) \implies 960 - 135d^2 = 448 - 7d^2$.
4. $512 = 128d^2 \implies d^2 = 4 \implies \mathbf{d = \pm 2}$. Taking $d = 2$, parts are: $2, 6, 10, 14$.

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Module 12 | Applied Contextual Modeling: Section E Case Studies

13.1 Case Study Archetype 1: The Semicircular Concentric Spiral

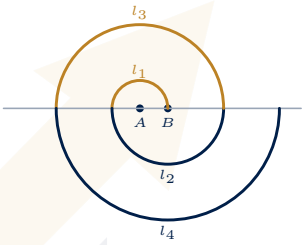
A spiral is made up of successive semicircles with alternating centers at A and B, of radii $r_1 = 0.5$ cm, $r_2 = 1.0$ cm, $r_3 = 1.5$ cm, ...

Mathematical Deconstruction

- Semicircle length: $l_k = \pi r_k \implies l_1 = 0.5\pi, l_2 = 1.0\pi, \ldots$
- Forms an AP with $a = 0.5\pi$ and $d = 0.5\pi$.
- **Total Length of 13 Semicircles ($n = 13$):**

$$S_{13} = \frac{13}{2}[2(0.5\pi) + 12(0.5\pi)] = \frac{13}{2}[7\pi] = \frac{91}{2}\pi$$

Using $\pi = \frac{22}{7}$: $S_{13} = \frac{91}{2} \times \frac{22}{7} = \mathbf{143}$ cm.



13.2 Case Study Archetype 2: The Potato Race / Retrieval Mechanics

Case Study Model | The Potato Race Retrieval Mechanics

Context: A bucket is placed 5 m from the 1st potato. Other 9 potatoes are spaced 3 m apart in a line (10 potatoes total). A competitor runs from the bucket, picks up each potato, and returns it to the bucket.

Step-by-Step Resolution:

1. Distance for 1st potato $= 2 \times 5 = \mathbf{10}$ m. Distance for 2nd potato $= 2 \times (5 + 3) = \mathbf{16}$ m.
2. Distances form an AP: $\mathbf{10}, \mathbf{16}, \mathbf{22}, \ldots$ with $a = 10, d = 6$, and total terms $n = 10$.
3. **Total Distance Run:** $S_{10} = \frac{10}{2}[2(10) + (10 - 1)6] = 5[20 + 54] = 5[74] = \mathbf{370}$ meters.

13.3 Case Study Archetype 3: Production Scheduling & Financial Installments

A manufacturer produced 600 cars in Year 3 and 700 cars in Year 7 with uniform annual growth:

- $a + 2d = 600$ and $a + 6d = 700 \implies 4d = 100 \implies \mathbf{d = 25}$ cars/year, First year: $a = \mathbf{550}$ cars.
- Production in Year 10: $a_{10} = a + 9d = 550 + 9(25) = \mathbf{775}$ cars.
- Total production in first 7 years: $S_7 = \frac{7}{2}[550 + 700] = 7 \times 625 = \mathbf{4375}$ cars.

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ARITHMETIC PROGRESSIONS

Module 13 | Board Traps, Verification Hacks & 5-Point Audit

14.1 The Top 8 Arithmetic Progression Board Traps

Trap 1: The Non-Integer Term Trap ($n \notin \mathbb{N}$): Accepting fractional or negative values for n. Position index must strictly be a counting number: $\mathbf{n \in \{1, 2, 3, \dots\}}$. If $n = 23.4$, the term does NOT exist!

Trap 2: The "Between" vs "From" Interval Trap: Numbers *"between* 100 *and* 500*"* strictly **excludes** 100 and 500. Numbers *"from* 100 *to* 500*"* **includes** 100 and 500 (if divisible).

Trap 3: The Common Difference Sign Reversal Trap: In a decreasing AP $(20, 16, 12, \dots)$, writing $d = 4$ instead of $\mathbf{d = -4}$. This completely reverses the signs of subsequent calculations!

Trap 4: The Symmetrical 4-Term Common Difference Error: Selecting terms as $(a - 3d, a - d, a + d, a + 3d)$ and assuming the common difference is d. The actual common difference is **2d**.

Trap 5: The Boundary Blindness Trap in $a_n = S_n - S_{n-1}$: Attempting to calculate a_1 as $S_1 - S_0$. S_0 has no mathematical meaning in Class 10 AP. Always set $\mathbf{a_1 = S_1}$ directly!

Trap 6: The Distributive Factor Error in S_n: Forgetting to distribute $\frac{n}{2}$ to both terms: writing $\frac{n}{2}[2a + (n-1)d]$ as $na + (n-1)d$. Always preserve the outer brackets!

Trap 7: The First Negative Term Inequality Sign Reversal: In $-4n < -125$, forgetting to reverse the inequality sign when dividing by -4: writing $n < 31.25$ instead of $\mathbf{n > 31.25}$.

Trap 8: Confusing Position (n) with Value (a_n): Mixing up the question *"Which term is* 78*?"* ($n =$?) with *"Find the* 78*-th term"* ($a_{78} =$?).

14.2 High-Speed Verification Hacks

Traditional Slow Check (High Risk of Calculation Fatigue)	The Maths Manthan Instant Verification Hack
Verifying whether $S_n = 3n^2 + 5n$ yields an AP: Compute S_1, S_2, S_3, S_4. Find $a_1 = S_1, a_2 = S_2 - S_1, a_3 = S_3 - S_2$. Check if $a_3 - a_2 = a_2 - a_1$. Time taken: ≈ 3 minutes. High risk of basic subtraction slip.	Look at the polynomial coefficients of $S_n = An^2 + Bn$: $\mathbf{d = 2A = 2(3) = 6}$ $\mathbf{a = A + B = 3 + 5 = 8}$ Instant check takes **3 seconds** with 100% algebraic certainty!

14.3 The 5-Point Board Examination Pre-Submission Checklist

Final 60-Second Answer Sheet Verification Protocol:

- ☐ **Domain Integrity Stated:** Did you verify that $n \in \mathbb{N}$ before accepting any value of n?
- ☐ **Symbolic Formula Declared:** Did you write $a_n = a + (n-1)d$ or $S_n = \frac{n}{2}[2a + (n-1)d]$ before substituting numeric values?
- ☐ **Sign of d Verified:** If the series was decreasing, did you ensure d was negative throughout?
- ☐ **Double Answer Justified:** For questions with two roots for n, did you explicitly explain that the intermediate terms sum to zero?
- ☐ **Physical Units Attached:** Are word problems and case studies concluded with correct physical units (meters, cars, cm)?

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